AI IMAGE FORENSICS SUITE

Complete Codebase & File Navigation Guide for Developers

Generated: August 2026 • Architecture: Next.js 15 + FastAPI + Multi-Scale 2D-FFT + Vision LLM

PROJECT OVERVIEW

This repository contains a high-precision digital image forensics suite engineered to distinguish authentic optical photographs from synthetic AI generations (Midjourney, Stable Diffusion, Flux, DALL-E, Firefly). It combines mathematical frequency-domain transforms, hardware EXIF/C2PA metadata provenance, deep convolutional features, and conversational AI forensics.

1. Core Scientific & AI Detection Engines (Backend Services)

`backend/app/services/frequency_analysis.py` — Mathematical 2D-FFT & Texture Engine

Key Mechanisms:

- 48-band radial frequency power spectrum fitting for natural 1/f^alpha optical decay slope.
- Azimuthal anisotropy index (16 angular sectors) detecting lattice harmonics from transposed convolutions.
- Laplacian noise residual variance evaluating physical Poisson photon noise vs Gaussian denoising.
- Local Binary Pattern (LBP) entropy, Canny edge contour coherence & color histogram smoothness.

Why You Must Review It: Contains the core mathematical algorithms that expose invisible generative artifacts in frequency space.

`backend/app/services/ensemble.py` — Dynamic Confidence-Weighted Voting Engine

Key Mechanisms:

- Dynamic margin formula: Effective Weight = Base Weight * max(2 * |score - 0.5|, 0.20).
- Integrates spectral (40%), convolutional CNN (45%), and vision LLM (15%) signals.
- Calibrated decision threshold at 0.52 to eliminate ambiguous grey-zone verdicts.

Why You Must Review It: Determines how multi-modal detection signals are mathematically blended into the final verdict.

`backend/app/services/metadata_extractor.py` — Hardware EXIF & C2PA Provenance Inspector

Key Mechanisms:

- Detects 21 AI generation software signatures (Midjourney, ComfyUI, Automatic1111, DALL-E, Firefly).
- Hardware sensor whitelist with 17 camera manufacturers (Canon, Nikon, Sony, Apple, Leica).
- Binary C2PA manifest scanner (JUMBF chunks) & PNG prompt/workflow parameter chunk extraction.
- Calculates missing-EXIF heuristic suspicion penalty (+0.35).

Why You Must Review It: Provides hard cryptographic and hardware provenance attribution before relying on probabilistic models.

`backend/app/services/gemini_detector.py` — Vision LLM & SON AI Conversational Copilot

Key Mechanisms:

- 9-dimension chain-of-thought prompt (ray-tracing, eye catchlights, anatomy, micro-textures).
- Real-world entity fact-checking (verifying real physical existence of landmarks/species).
- Deterministic local forensic synthesis fallback allowing SON AI to answer offline with zero cloud API keys.

Why You Must Review It: Powers SON AI reasoning, high-level semantic validation, and responsive forensic Q&A explanations.

`backend/app/services/model_inference.py` — ONNX Deep CNN Runtime Engine

Key Mechanisms:
- Loads optimized ONNX visual classifier weights for offline neural evaluation.
- Standardized RGB normalization, tensor resizing, and high-speed CPU execution.

Why You Must Review It: Handles the local visual neural network inference pipeline.

2. API Endpoints & Request Orchestration

`backend/app/api/routes/detect.py` — Multi-Scale Inspection & Q&A Routes

Key Mechanisms:
- POST /api/v1/detect/image: Runs multi-scale scans across 3 resolutions (256px, 384px, 512px).
- POST /api/v1/detect/question: Feeds laboratory telemetry to SON AI for structured diagnostic answers.
- Safe multipart upload decoding, MIME sniffing, and human-readable confidence classification.

Why You Must Review It: The primary API controller coordinating all forensic layers, background threads, and client responses.

`backend/app/main.py` — FastAPI Application Initialization

Key Mechanisms:
- Initializes FastAPI app, CORS origin whitelist (http://localhost:3000), and rate limiting.
- Registers detection, auth, and health check route handlers (/health, /api/v1/detect).

Why You Must Review It: The root backend application entrypoint and middleware assembly hub.

`backend/app/api/routes/auth.py` — Authentication & User Account Security

Key Mechanisms:
- User registration, login, and profile retrieval.
- PBKDF2-SHA256 password hashing and signed JWT bearer token issuance.

Why You Must Review It: Protects forensic investigator accounts, workspace persistence, and user authentication.

3. Security Hardening & Zero-Billing Cost Guards

`backend/app/core/billing_guard.py` — Zero-Billing Protection & SHA-256 Cache

Key Mechanisms:
- SHA-256 binary hash caching: Identical image scans are served in <1ms for $0 API/compute cost.
- Daily external API quota enforcement to guarantee zero unexpected cloud billing spikes.

Why You Must Review It: Ensures the platform can be hosted publicly without vulnerability to runaway cloud bills.

`backend/app/core/security_middleware.py` — Anti-Abuse, Rate-Limiting & SSRF Defense

Key Mechanisms:
- Server-Side Request Forgery (SSRF) defense preventing probes of private/loopback IP addresses.
- Real file magic-byte inspection (libmagic) preventing polyglot file execution.
- IP-based sliding-window rate limiting & brute-force lockout mechanisms.

Why You Must Review It: Hardens the backend against DDoS, unauthorized scraping, and malicious file payload attacks.

4. Frontend UI & Interactive Forensic Workspace (Next.js 15)

`frontend/src/app/page.tsx` — Main Forensic Dashboard Orchestrator

Key Mechanisms:

- Manages global specimen state, active analysis tabs, and history navigation.
- Binds file upload handlers, comparator switches, and floating SON AI assistant.

Why You Must Review It: The root client-side page connecting all interactive tools, modals, and workspaces.

`frontend/src/components/ForensicsWorkspace.tsx` — Single Specimen Deep Forensic Workspace

Key Mechanisms:

- Interactive zoom loupe with pixel-level inspection.
- Simulated 2D-FFT Fourier canvas, radial power spectrum charts, and signal dials.
- Triggers instant client-side PDF certificate export via jsPDF.

Why You Must Review It: The core inspection interface where forensic investigators examine uploaded evidence.

`frontend/src/components/SideBySideComparator.tsx` — Dual Specimen Comparator (Channel A vs B)

Key Mechanisms:

- Original dual-column comparison layout with center VS divider.
- Synchronized dual 2D-FFT spectrogram canvases and differential metric comparison matrix.
- Instant benchmark test presets (AI Cyberpunk vs Authentic Golden Gate).

Why You Must Review It: Allows investigators to compare two images side-by-side to cross-examine frequency divergences.

`frontend/src/components/ImageAssistant.tsx` — SON AI Interactive Forensic Copilot

Key Mechanisms:

- Floating conversational drawer with structured Markdown rendering (headings, bullet points, bolding).
- Context-aware: Passes diagnostic telemetry (verdict, slope, anisotropy, camera) into questions.
- Pre-populated forensic prompt chips (e.g., "Why is this AI?", "Check Fourier decay").

Why You Must Review It: The user-facing chat assistant explaining scientific findings in clear, conversational language.

`frontend/src/components/UploadDropzone.tsx` — Resilient Drag-and-Drop Uploader

Key Mechanisms:

- Dual validation: Checks both MIME type and file extension (.jpg, .jpeg, .png, .webp, .tiff).
- Provides instant pre-upload size warnings and guidance for optimal analysis.

Why You Must Review It: The primary file ingestion gateway ensuring smooth, reliable image uploads.

`frontend/src/components/ForensicCertificateModal.tsx` — Printable Certificate of Authenticity

Key Mechanisms:

- Tamper-evident verification certificate with SHA-256 cryptographic seal.
- Formatted for high-resolution print and courtroom evidence export.

Why You Must Review It: Renders official verification certificates for legal, journalistic, or institutional records.

`frontend/src/lib/imageOptimizer.ts` — In-Browser Image Pre-Processor

Key Mechanisms:

- Downsamples massive RAW images client-side before network transfer.
- Friendly diagnostic error mapper translating HTTP error codes into plain guidance.

Why You Must Review It: Prevents network timeouts and server memory exhaustion while preserving high-frequency edges.

`frontend/src/app/globals.css` — **Editorial Design System & Dark/Light Palettes**

Key Mechanisms:

- Custom typography hierarchy (Georgia serif titles, Monospace telemetry badges).
- Side-by-side dual channel cards, signal glow meters, and print media rules.

Why You Must Review It: Defines the polished, professional visual identity of the entire platform.

5. Automated Test Suite (Validation & Quality Assurance)

`backend/tests/ (24 Automated Tests Passing)` — **Full Regression & Scientific Test Suite**

Key Mechanisms:

- test_spectral_accuracy.py: Verifies 1/f^alpha natural optical curves and artificial grid detection.
- test_detect.py: Tests multipart uploads, corrupt byte resilience, and size boundaries.
- test_gemini_detector.py: Validates dynamic ensemble voting and SON AI Q&A formatting.
- test_security_and_billing.py: Verifies SSRF protection, SHA-256 caching, and rate limiting.
- test_auth.py: Tests password hashing, JWT token creation, and auth flows.

Why You Must Review It: Run `pytest backend/tests -v` anytime to verify 100% test passing across the entire platform.

6. Recommended 3-Step Reading Pathway for Developers

Step 1: Understand the Science (Backend)

Read frequency_analysis.py -> metadata_extractor.py -> ensemble.py -> gemini_detector.py

Step 2: Understand the Pipeline & Security

Read detect.py -> billing_guard.py -> security_middleware.py

Step 3: Understand the User Experience (Frontend)

Read page.tsx -> ForensicsWorkspace.tsx -> SideBySideComparator.tsx -> ImageAssistant.tsx